

How to Wire the Stepper Motors and an Encoder

- Wire Lengths:
 - Switch: >6inches
 - Steering Encoder: >7 inches
 - Steering Motor; >4 inches
 - Left Motor: > 5 inches
 - Right Motor: >5 inches
- Wire Each Stepper Motor:
 - Use stepper motor datasheets or a multimeter to identify coil pairs.
Connect each motor's A+ A- and B+ B- wires to the correct driver terminals.
Keep wire lengths manageable and avoid crossing wires when possible.
- Connect the Encoders:
 - Use the 3-pin encoder for steering and the 5-pin encoders coming out of the stepper for the drive steppers.
 - When soldering the steering encoder there are 2 ways to make the encoder work correctly.
 1. Solder a wire between VCC and GP0.OR
 2. Remove resistor 4.
- Secure Your Wiring:
 - Make sure the steering encoder is attached to the bottom cover and routed through the vehicle before attaching the bottom cover. Label both ends of each wire for easy debugging later.
- Stay Safe:
 - Disconnect power before making any changes.
 - Avoid touching stepper drivers when powered—they get hot quickly.
 - Keep loose wires away from fan blades or moving parts.

Reminders for Assembling the Car

- Check All Parts First:
 - Lay out every printed piece and compare against your assembly plan or parts list.
 - Verify that parts are cleanly printed, supports are removed, and undamaged before starting.
- Check tolerances:
 - Lightly sand mating surfaces if parts are too tight or warped.
- Use Adhesives:
 - Superglue (CA glue) for plastic interfaces.
 - Loctite 425 for metal-to-metal interfaces.
 - Apply sparingly — a little goes a long way and avoids mess.
- Check Alignment and Motion:
 - Spin wheels to ensure smooth rotation and even clearance.
 - Test steering movement before locking everything in place.
 - Steering should tighten until the shoulder bolt almost peeks out from the nylon insert in the Nylock nut.
- Finish and Detail (Optional):
 - Sand seams, then prime and paint if desired.
 - Apply decals or stickers after clear-coating for best results.
- Stay Safe:
 - Wear eye protection and gloves when cutting, sanding, or using adhesives.
 - Work in a well-ventilated area when using glue or paint fumes.

3DP Parts List

3D Prints, Recommended Materials, and Quantity		
Parts	Filament	Qty
Main Body	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
Bottom Cover	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
Front Bumper	Bambu 95A TPU, Markforged TPU	1
Rear Bumper	Bambu 95A TPU, Markforged TPU	1
Sensor Plate	Bambu PETG-CF, Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
Camera Mount	Bambu PETG-CF, Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
Wheel Hub	Bambu PETG-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	2
Wheel Hub Stub	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	2
Wheel Hub Washer	Bambu PETG-CF, Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	3
Steering Washer	Bambu PETG-CF, Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
Steering Rod	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
Motor Connector 1	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
Motor Connector 2	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
12t Pinion	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
55t Gear	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
Pinion Clamp	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
Steering dead axle	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
Stepper Spacer	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
Front Wheel Rim	Bambu PETG-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	2
Rear Wheel	Bambu PETG-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	2
Tire	Bambu 95A TPU, Markforged TPU	4
Front Right Shroud	Bambu PETG, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
Front Left Shroud	Bambu PETG, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
Rear Right Shroud	Bambu PETG, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
Rear Left Shroud	Bambu PETG, Polymaker HT PLA, Bambu PPA-CF, Siraya Tech PPA-CF	1
Tire	Bambu 95A TPU, Markforged TPU	4